# Variant-Informed Pipeline for Evidence-based drug target Ranking- ViP targets

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## MOTIVATION

***Background***

- LLM agents can generate oncology hypotheses at scale, but checking them against evidence still takes weeks of expert curation.
- Multi-agent systems can propose, challenge, and revise hypotheses across omics data and literature.

***Challenges***

- LLM outputs contain unsupported claims, and holistic scoring can't be trusted.
- Controlled-access data can't go to cloud models, so inference must run locally with open-weight LLMs on HPC.
- Deterministic omics steps and LLM judgment steps must work together reproducibly.
- Manual review doesn't scale and doesn't support reanalysis as new evidence appears.

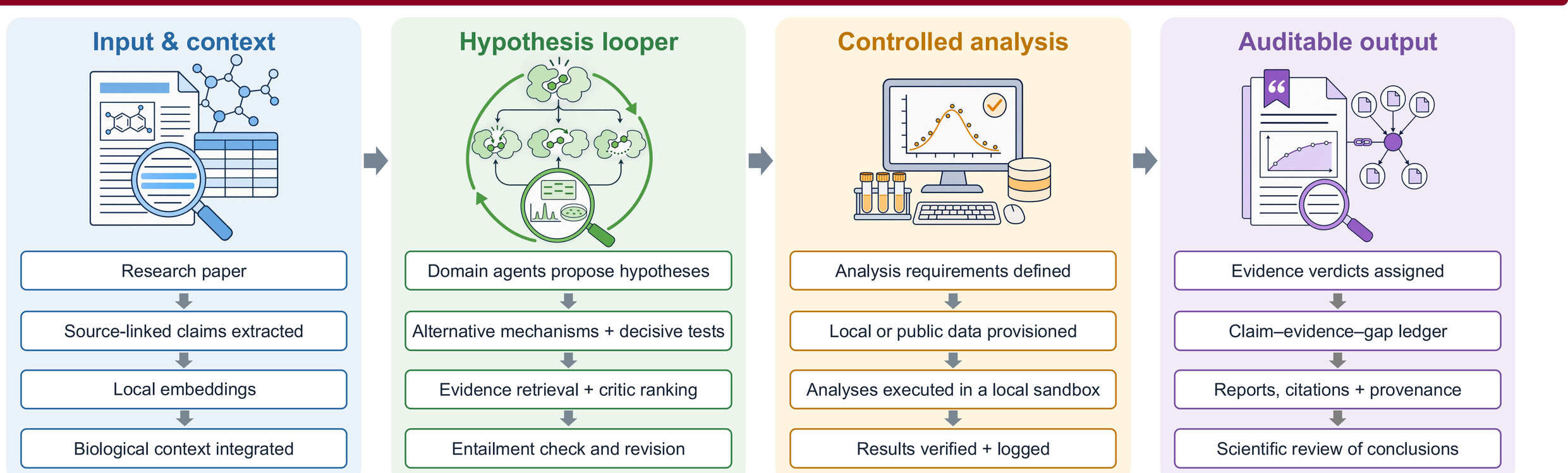
***Goals***

- We built OnKoLoop, for multi-agent search and validation of alternative hypotheses.
- It has the following parts:
  - Multi-agent hypothesis generation.
  - Structured claim extraction.
  - Triage with an alternative-hypothesis ledger and leave-one-evidence-out attribution.
  - Provenance-tagged decision records.

***Translational impact***

- Candidates reach the bench with their supporting evidence and alternative explanations attached with the audit trails logged.
- Auditable, evidence-linked recommendations with systematic reanalysis as new evidence appears.

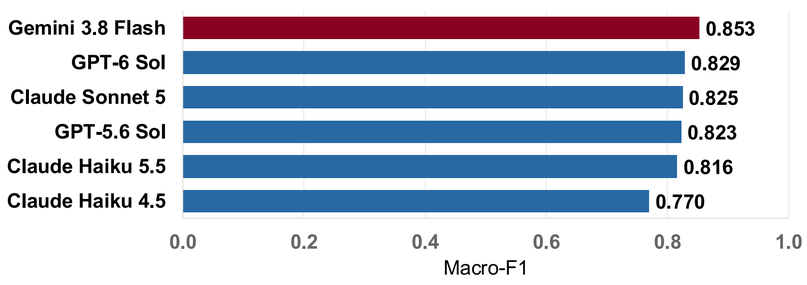
## APPROACH



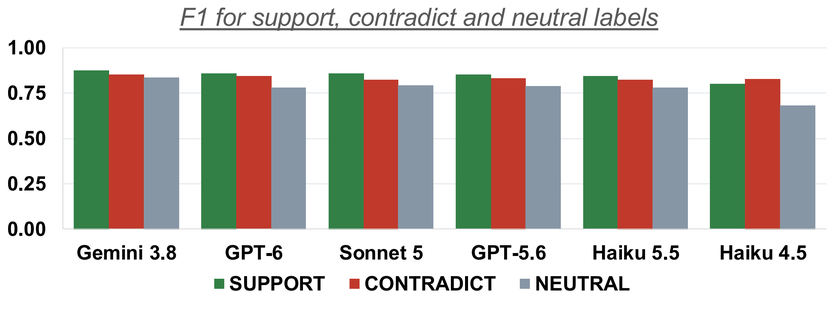

## RESULTS

### SCIENTIFIC CLAIM VERIFICATION TO VALIDATE THE METHOD

*SciFact claim–document pairs across six evaluated model configurations establishes confidence in the agentic system*




*F1 for support, contradict and neutral labels*

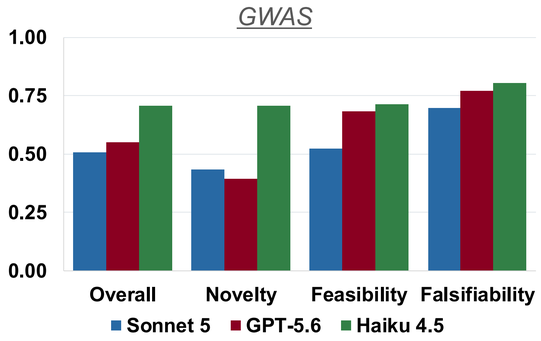



- Gemini 3.8 Flash highest scores also as the judge LLM is the same model. We are looking at other ways to score it.
- Majority-class accuracy: 0.393. Scores describe claim-to-document classification.

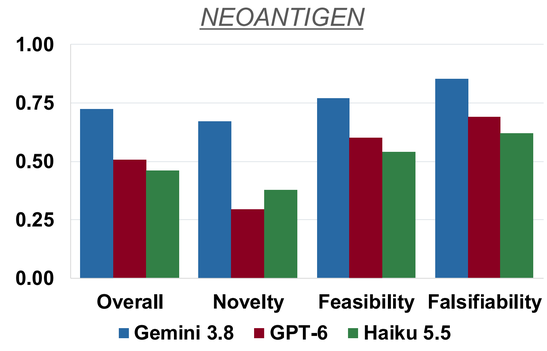
### INTERNAL MODELS VALIDATION OF THE HARNESS

*ARGO MODELS*

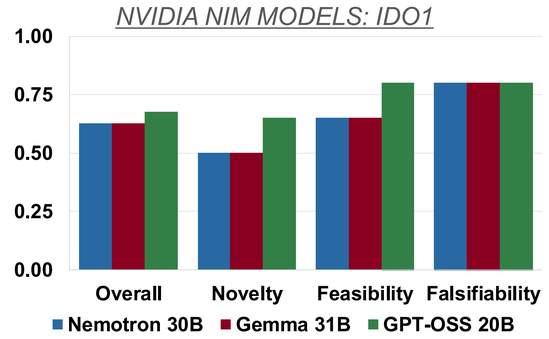
*GWAS*




*NEOANTIGEN*

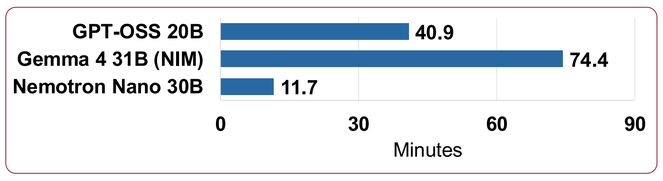



*NVIDIA NIM MODELS: IDO1*

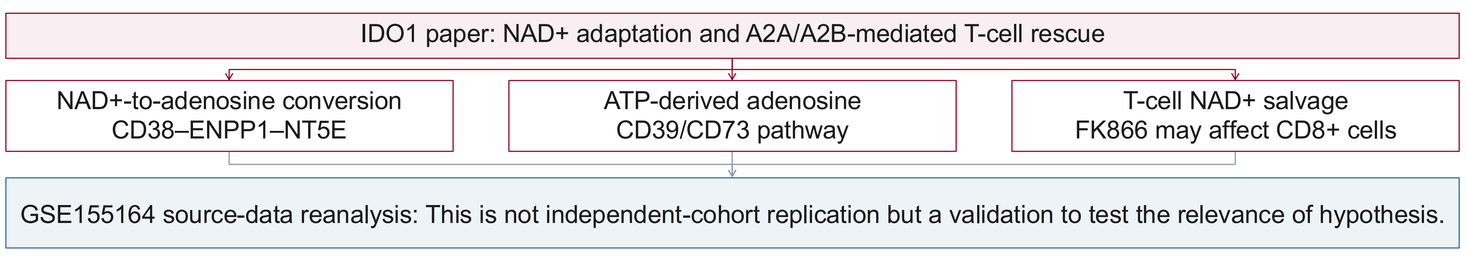



- Experiments use different papers, judges and configurations; profiles stay grouped.
- Nemotron self-judged its NIM run.
- Now since, the harness works we tried the models on locally hosted models.

*Generation time*




### EVIDENCE TRAIL AND GENERATION TIME




Representative hypotheses.
RNA proxies do not establish metabolite flux. FK866/CD8 evidence in diabetic mice is cross-context support only (PMID 42243084).

**TO KNOW MORE SCAN**

Recent updates about the code, manuscript and others will be available here or email: suranap@uchicago.edu, madduri@anl.gov

## FUTURE WORK

- OnKoLoop linked hypotheses to evidence and proposed tests.
- Validate the IDO1 hypotheses experimentally with collaborators.
- Complete fully local inference so sensitive data never leaves the lab and compare local models against hosted ones.

*Code will be released with the preprint.*

***OnKoLoop can do hypothesis generation and validation in a local environment for cancer research.***

## ACKNOWLEDGEMENTS

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